

BENCHMARK: ACCURATE — fit consistent with published values

Results

Mid-Transit Time (Tmid)	2458605.8092 +/- 0.0024 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.155 +/- 0.012
Transit depth (Rp/Rs)^2	2.4 +/- 0.39 [%]
Orbital Inclination (inc)	89.93 +/- 0.67
Airmass coefficient 1 (a1)	1.396 +/- 0.021
Airmass coefficient 2 (a2)	-0.305 +/- 0.033
Scatter in the residuals of the lightcurve fit is	1.53 %
Best Comparison Star	None
Optimal Aperture	3.75
Optimal Annulus	10.77
Transit Duration (day)	0.0999 +/- 0.0014

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.155 ± 0.012 vs 0.1406 (deviation 0.82σ)
Duration fit vs published	0.0999 d vs 0.0975 d (deviation 1.17σ)
Mid-time O–C	-2.1 min
Depth SNR	8.8
Residual scatter	1.53 %

Light curve

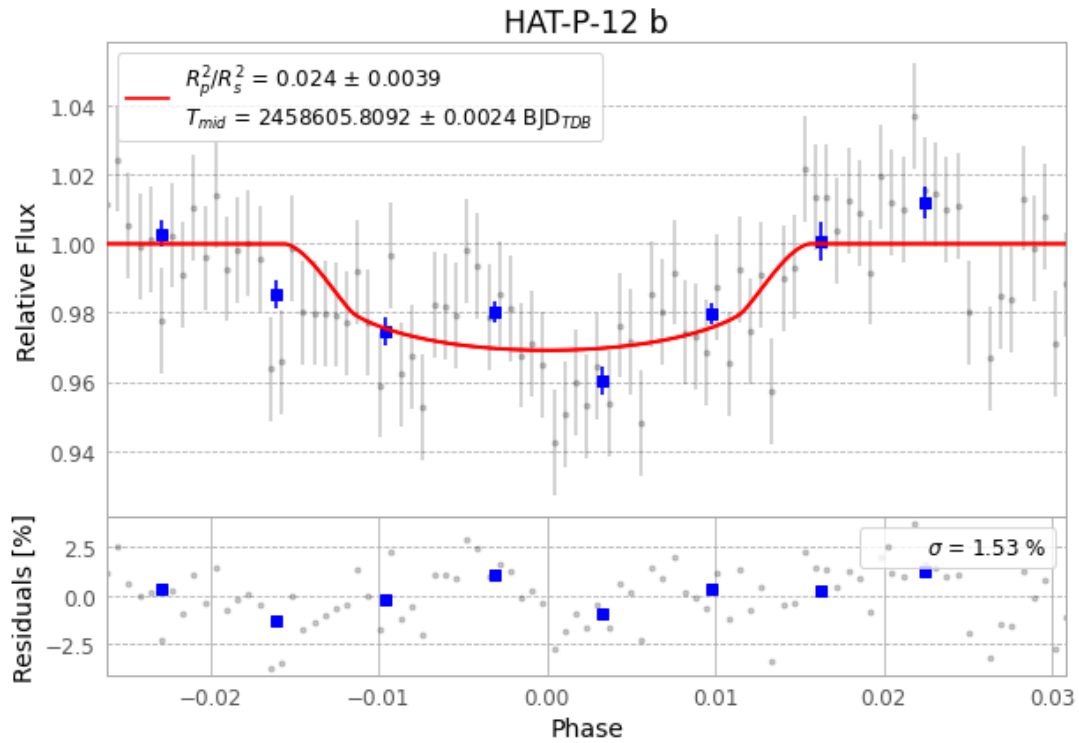


Figure 1 — FinalLightCurve_HAT-P-12 b_2019-05-01.png (SHA-256 2ab118f02f467e06...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [422, 173], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 3cefe8858bbf53bb...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b

Data provenance

89 FITS frames (58 MB), HATP-12190502051814.FITS → HATP-12190502094212.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-HATP-12-190502.log (457a121a2f4b...), FinalLightCurve_HAT-P-12 b_2019-05-01.png (2ab118f02f46...), FinalLightCurve_HAT-P-12 b_2019-05-01.csv (207de4d78494...)

Compute resources

Wall time 120.3 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-01 08:02Z.